CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (CL) Examination

May 2013

CL 403 Design of Structures- I

Date: 09.05.2013, Thursday | **Time: 10:00 a.m. To 01:00 p.m.** | **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Select the most appropriate correct answer from the given options for the following multiple-choice questions and write in the main answer book. **[02]**

1. The ultimate strain in concrete at the outermost compression fiber in bending as per limit state theory is
   a) 0.002 b) 0.035 c) 0.0035 d) 0.446
2. When assessing the strength of a structure for limit state of collapse, the partial safety factor for concrete should be taken as
   a) 1.5 b) 1.05 c) 1.55 d) 1.15

Q - 2 (a) A simply supported RC beam 380 mm wide and 750 mm deep, carries a uniformly distributed load of 80 kN/m (including self-weight) over a span of 6 m. The beam is reinforced with 6 nos. 22 mm diameter bars of grade Fe 500 on tension face. Design and detail the shear reinforcement. Also check this beam for deflection. **[09]**

(a) Using inclined stirrups and no curtailment of bars.

(b) Using bent-up bars.

Use M 25 and effective cover 50 mm, width of support is 380 mm.

(b) Design and detail a RC slab for a room having inside dimensions 3.5 m × 7 m. The thickness of supporting wall is 300 mm. The slab carries 75 mm thick lime concrete at its top, the unit weight of which may be taken as 20 kN/m^3. The live load on the slab is taken as 2.5 kN/m^2. Assume the slab to be simply supported at the ends. Use M 20 grade concrete and Fe 415 steel. **[09]**

OR

Q - 2 (a) Design and detail the simply supported flanged beam for following data: **[09]**
D_f = 100 mm, D =750 mm, b_w = 350 mm, spacing of beams = 3500 mm c/c, effective span = 10 m, cover = 90 mm, d = 660 mm and imposed loads = 10 kN/m^2. Use M 20 & Fe 415.

(b) A drawing room of a residential building measures 3.8 m × 6 m. It is supported on 350 mm thick walls on all four sides. The slab is simply supported at edges with no provision to resist torsion at corners. Design the slab using M 25 grade concrete and Fe 415 steel. **[09]**

Q - 3 (a) Design and detail the reinforcement for a short column for the following data: **[09]**
Column size : 350 mm × 550 mm , P_u : 1800 kN, M_{ux} : 140 kNm, M_{uy} 100 kNm. Use M25 and Fe415.

(b) A solid footing has to transfer a dead load of 900 kN and an imposed load of 300 kN from a square column 350 mm × 450 mm with 16 mm diameter bars. Design the isolated uniform depth footing. Soil bearing capacity of soil is 180 kN/m^2. Use M 20 grade concrete and Fe 415 steel. **[06]**

OR

Q - 3 (a) A R.C. short column square in section has to resist a factored axial load of 1700 kN. Use M20 and Fe415. Assuming 1.8% of longitudinal steel reinforcement. Determine (a) size of square column, (b) area of longitudinal reinforcement, (c) pitch and diameter of lateral ties. Draw sketch of the sectional elevation and plan. [09]

(b) A solid footing has to transfer a dead load of 1000 kN and an imposed load of 500 kN from a rectangular column 300 mm × 600 mm. Design the isolated uniform depth footing. Soil bearing capacity of soil is 200 kN/m^2. Use M 25 grade concrete and Fe 415 steel. [06]

SECTION – II

Q - 4 (a) [03]

1. The ratio of the effective throat of a fillet weld to its size is
   a) 0.707 b) Less than or equal 0.707 c) Less than 1.0 d) More than 0.707
2. The maximum effective size of a fillet admissible at the rounded edges is
   a) 0.75 times the thickness of the plate b) Equal to the thickness of the plate
   b) Half of any thickness of plate d) Usually more than the thickness of the plate
3. The ratio of the strength of the weld material to that of the parent body is usually
   a) More than one b) Less than one c) Equal to one d) Equal or more than one

(b) Describe advantages and disadvantages of bolted connections. [03]

Q - 5 (a) Two plates 100 mm wide and 12 mm,20 mm thick are connected by lap joint to resist tensile load of 100 kN. Design a lap joint using M12 bolts of grade 4.6 and grade 410 plates as well as design the connection using shop welding. [05]

(b) An unequal angle 1.5m long, of a truss is connected to the gusset plate. It carries ultimate tension load of 230 kN. Design the section using 6mm weld. Assume f_y= 250 Mpa.and f_u= 410 Mpa. [10]

OR

Q - 5 (a) Determine the shear and tension capacity of 16 mm ø bolt of grade 8.8 connecting 12mm plate. Also determine the strength of 6mm fillet weld.Full references attract maximum marks. [05]

(b) An unequal angle 1.5m long, of a truss is connected to the gusset plate. It carries ultimate tension load of 230 kN. Design the section using bolted connection. Assume f_y= 250 Mpa.and f_u= 410 Mpa. [10]

Q - 6 (a) Calculate compressive strength of ISA 75 × 75 × 6 mm assuming that the angle is loaded through one leg when, [07]

(i) It is connected by 1 bolt at each end.
(ii) It is connected by 2 bolts at each end.
(iii)It is wlded at each end.

The length of member is 2 m and fy = 250 MPa.

OR

(a) A column of 7m long effectively held in position and restrained against rotation at both ends.Design the column by other than quick oriented method to carry axial compresion of 1600kN. [07]

(b) Design a laterally unrestrained simply supported beam over a span of 5 m which carries a factored UDL of 50 kN. Carryout atleast one design check. Take f_y= 250 N/mm^2 [07]

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